14. Certified Weyl nodes and slice topology in a lattice model

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Energy bands in a three-dimensional crystal can meet at isolated points instead of remaining separated everywhere. Near certain crossings, the equations resemble those of a Weyl particle. Each crossing carries a handedness-like topological label. This problem asks whether we can find every such crossing in one lattice model, prove their labels, and connect them to the behavior of two-dimensional momentum slices.

Key terms

- **Semimetal** — A system in which the occupied and unoccupied electronic states are not separated by a full insulating gap; the selected model has isolated band crossings.
- **Weyl node** — A simple crossing of two bands with a characteristic linear structure and nonzero topological charge.
- **Chirality** — The signed handedness of a simple Weyl crossing, defined here from its velocity matrix.
- **Velocity matrix** — The derivatives describing how the Hamiltonian changes in each momentum direction near a node.
- **Berry flux** — The integrated Berry curvature through a closed surface around a crossing.
- **Momentum slice** — A two-dimensional section of the three-dimensional Brillouin zone, obtained by holding one momentum coordinate fixed.
- **Fermi arc** — An open surface-state contour whose relation to bulk nodes depends on the chosen surface and energy.
- **Root isolation** — A mathematical procedure proving that a small region contains a solution, often uniquely.
- **Completeness** — A proof that no additional nodes have been missed anywhere in the claimed domain.

Why this matters

Weyl nodes connect a local band crossing to a global topological constraint. They help explain unusual electronic structures and surface states, and provide controlled settings for studying the relation between particle-like excitations and lattice physics. Missing even a pair of nodes can change the interpretation of a phase diagram.

What progress would mean

A certified catalog would establish all crossings and slice invariants of the selected model. It would give a reliable starting point for surface and response calculations, without pretending that bulk

nodes uniquely determine every surface arc.

Core question

Can all nodes in a specified two-band three-dimensional Hamiltonian be isolated, assigned chirality, and related to the Chern numbers of gapped momentum slices?

Scope and assumptions

Begin with $H(k) = \sin k_x\, \sigma_x + \sin k_y\, \sigma_y + (2 - \cos k_x - \cos k_y - \cos k_z)\sigma_z$, k∈$\mathrm{T}^3$, lower band occupied away from nodes. Fix orientation and Pauli-matrix conventions. This exact benchmark has nodes at $(0,0,\pm\pi/2)$.

Subsequent searches must fix orbitals, hopping support, parameter box, and symmetry representation. Dirac points and nodal lines are separate extensions because their protection criteria differ.

Mathematical target

For H=d_0I+d·σ, nodes solve d(k)=0. A simple node has invertible velocity matrix $V_{ij} = \partial d_i/\partial k_j$. Define Weyl chirality χ=sgn det V, and explicitly relate this to the lower-band Chern flux under the chosen convention. Prove the complement of all isolating neighborhoods has |d(k)|>0.

For fixed k_z away from nodes, certify the two-dimensional occupied Chern number. Its jumps track node charge. A slab calculation requires a chosen surface normal and termination; arc details are not determined solely by bulk node positions.

Required outputs and proof obligations

Return all root enclosures, uniqueness/completeness proof, Jacobian signs, slice Chern numbers, and optional slab spectrum with error control. Nodes need not have closed-form coordinates: certified isolating boxes are acceptable. Robustness is scoped to perturbations that prevent annihilation with opposite total charge.

Validation and rejection controls

Verify the two benchmark roots analytically and opposite determinant signs. Check total charge on the periodic Brillouin zone vanishes. Confirm slice invariants only on gapped slices. Test a perturbation moving nodes and a parameter change permitting pair annihilation.

Milestones and research extension

Baseline: complete benchmark node and slice audit. Strong: certified parameter-dependent node motion or one bounded model family. Research extension: a symmetry-protected Dirac model with explicit little-group constraints, or nodal-link invariants with their own definitions.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Wan et al., Electronic Structure of Pyrochlore Iridates (published as Topological semimetal and Fermi-arc surface states) — Reference for Weyl nodes and surface arcs; benchmark and proof obligations here are explicitly specified.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$d(k) = (\sin k_x, \sin k_y, 2 - \cos k_x - \cos k_y - \cos k_z), \quad d(k_*) = 0,$$

$$k_* = (0, 0, \pm\pi/2), \qquad V_{ij} = \partial_{k_j} d_i(k_*), \qquad \chi = \operatorname{sgn}\det V.$$

The two determinants are $+1$ and -1 in the stated coordinate order. Relate this convention explicitly to the occupied-band flux. Certify absence of other nodes on the complement and the topology of gapped slices; surface arcs additionally depend on the chosen boundary.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Major — oversimplified symmetry (22–25).** Time reversal and inversion alone are not a universal recipe for a protected fourfold Dirac crossing; representation and additional symmetry constraints matter. They cannot replace an explicit model.
2. **Major — isotropy assumed (73–87).** A general Weyl cone has a velocity matrix, can be tilted, and is not specified by one scalar v_F. Its determinant gives the simple-node chirality in a declared convention.
3. **Major — completeness (99–105).** Finding roots and checking their charge sum does not prove there are no additional zero-charge pairs. A proof on the complement of the isolating boxes is needed.

4. **Major — boundary overclaim.** Fermi-arc connectivity depends on surface projection and termination; opposite nodes can project together. The revision uses slice topology as the robust intermediate claim and separates surface-specific predictions.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **Can all nodes in a specified two-band three-dimensional Hamiltonian be isolated, assigned chirality, and related to the Chern numbers of gapped momentum slices?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: complete benchmark node and slice audit. Strong: certified parameter-dependent node motion or one bounded model family. Research extension: a symmetry-protected Dirac model with explicit little-group constraints, or nodal-link invariants with their own definitions.

Sources supporting the corrections

- Wan et al., Electronic Structure of Pyrochlore Iridates (published as Topological semimetal and Fermi-arc surface states) — Reference for Weyl nodes and surface arcs; benchmark and proof obligations here are explicitly specified.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.

- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.